Bayesian Threshold Vector Autoregression

This repository provides an example of a Bayesian threshold vector autoregression (TVAR) for studying nonlinear and state-dependent macroeconomic effects of a government spending shock. The model allows both the dynamic relationships among the variables and the covariance matrix of the reduced-form shocks to differ between low- and high-utilization states.

Model Specification

Let the vector of endogenous variables be

$$Y_t = \begin{bmatrix} g_t & \tau_t & y_t & \widetilde{CU}_t \end{bmatrix}', \tag{1}$$

where

- g_t is the quarterly growth rate of real government spending;
- τ_t is the quarterly growth rate of real net taxes;
- y_t is the quarterly growth rate of real GDP; and
- $\widetilde{CU}_t$ is mean-adjusted capacity utilization.

The threshold variable is lagged mean-adjusted capacity utilization:

$$S_{t-d} = \widetilde{CU}_{t-d}, \tag{2}$$

where $d = 1$. Let γ denote the threshold value. In the baseline implementation, γ is fixed at the sample mean of the threshold variable. The regime indicator is therefore

$$r_t = \begin{cases} 1, & S_{t-d} \leq \gamma, \\ 2, & S_{t-d} > \gamma. \end{cases} \tag{3}$$

Regime 1 represents the low-utilization state, while Regime 2 represents the high-utilization state.

Conditional on the prevailing regime, the dynamics of Y_t are given by

$$Y_t = c_1 + \sum_{\ell=1}^{p} A_{1,\ell} Y_{t-\ell} + u_{1,t}, \qquad u_{1,t} \sim N(0, Q_1), \qquad \text{if } S_{t-d} \leq \gamma, \tag{4}$$

and

$$Y_t = c_2 + \sum_{\ell=1}^{p} A_{2,\ell} Y_{t-\ell} + u_{2,t}, \qquad u_{2,t} \sim N(0, Q_2), \qquad \text{if } S_{t-d} > \gamma. \tag{5}$$

The implementation uses $p = 4$ quarterly lags and includes a constant. The model therefore allows

$$\Phi_1 \neq \Phi_2 \qquad \text{and} \qquad Q_1 \neq Q_2, \tag{6}$$

where

$$\Phi_r = \begin{bmatrix} c_r & A_{r,1} & A_{r,2} & \cdots & A_{r,p} \end{bmatrix}' \tag{7}$$

contains the coefficients for regime r. Thus, both the propagation of shocks and the variance and contemporaneous correlation of the shocks may differ across utilization states.

Although the model is estimated separately for the two regimes, the lagged regressors retain their observed historical values. Therefore, an observation assigned to one regime may contain lagged variables that were generated while the economy was in the other regime.

Regime-Specific Regression Form

For each regime $r \in \{1, 2\}$, let Y_r collect the observations assigned to regime r, and let Z_r contain the corresponding regressors:

$$Z_t = \begin{bmatrix} 1 & Y'_{t-1} & Y'_{t-2} & \cdots & Y'_{t-p} \end{bmatrix}. \tag{8}$$

The regime-specific model can then be written as

$$Y_r = Z_r \Phi_r + U_r, \tag{9}$$

where

$$U_r \sim MN(0, I_{T_r}, Q_r), \tag{10}$$

and T_r is the number of observations assigned to regime r.

Vectorizing the model gives

$$y_r = (I_N \otimes Z_r)\, \phi_r + u_r, \tag{11}$$

where

$$y_r = \mathrm{vec}(Y_r), \qquad \phi_r = \mathrm{vec}(\Phi_r), \qquad u_r = \mathrm{vec}(U_r). \tag{12}$$

Bayesian Estimation

The parameters of the two regimes are treated as separate random variables. The prior distributions are assumed to be independent across regimes:

$$p(\phi_1, Q_1, \phi_2, Q_2) = p(\phi_1)p(Q_1)p(\phi_2)p(Q_2). \tag{13}$$

For each regime, the prior for the VAR coefficients is normal:

$$\phi_r \sim N(\widetilde{\phi}_{0,r}, H_r), \tag{14}$$

where $\widetilde{\phi}_{0,r}$ is the prior mean and H_r is the prior covariance matrix.

Because government spending, net taxes, and GDP enter as growth rates and capacity utilization is mean-adjusted, the prior means of the lag coefficients are set equal to zero:

$$\widetilde{\phi}_{0,r} = 0. \tag{15}$$

The covariance matrix in each regime is assigned an inverse-Wishart prior:

$$Q_r \sim \mathcal{IW}(Q_{0,r}, \alpha_{0,r}), \tag{16}$$

where $Q_{0,r}$ is the prior scale matrix and $\alpha_{0,r}$ is the prior degrees-of-freedom parameter.

Minnesota Prior

A Minnesota-style prior is used for the regime-specific VAR coefficients. Let $a_{r,ij,\ell}$ denote the coefficient on lag ℓ of variable j in equation i and regime r. Its prior variance is

$$\mathrm{Var}(a_{r,ij,\ell}) = \begin{cases} \dfrac{\lambda_1^2}{\ell^{2\lambda_3}}, & i = j, \\ \dfrac{\lambda_1^2 \lambda_2^2}{\ell^{2\lambda_3}} \dfrac{\widehat{\sigma}_{r,i}^2}{\widehat{\sigma}_{r,j}^2}, & i \neq j, \end{cases} \tag{17}$$

where $\widehat{\sigma}_{r,i}^2$ is the residual variance from a univariate autoregression for variable i, estimated using observations from regime r.

The prior variance for the constant in equation i is

$$\mathrm{Var}(c_{r,i}) = \lambda_4^2 \widehat{\sigma}_{r,i}^2. \tag{18}$$

The hyperparameters are set to

$$\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 1. \tag{19}$$

Under this specification, $\lambda_1 = 1$ determines the overall prior tightness. Setting $\lambda_2 = 1$ means that cross-variable lags receive no additional shrinkage beyond the adjustment for differences in variable scale. Because $\lambda_3 = 1$, the prior variance declines with the square of the lag length, so higher-order lags are more strongly shrunk toward zero. Finally, $\lambda_4 = 1$ determines the prior variance placed on the regime-specific constant terms.

These hyperparameters are used as a baseline prior specification rather than being selected to reproduce the results of a particular empirical study.

Conditional Posterior Distributions

Conditional on Q_r, the posterior distribution of the coefficients in regime r is normal:

$$\phi_r \mid Q_r, Y_r \sim N(\phi_r^*, H_r^*), \tag{20}$$

where

$$H_r^* = \left[H_r^{-1} + Q_r^{-1} \otimes (Z_r' Z_r) \right]^{-1}, \tag{21}$$

and

$$\phi_r^* = H_r^* \left[H_r^{-1} \widetilde{\phi}_{0,r} + \left(Q_r^{-1} \otimes Z_r' \right) y_r \right]. \tag{22}$$

Conditional on Φ_r, the posterior distribution of the covariance matrix is inverse Wishart:

$$Q_r \mid \Phi_r, Y_r \sim \mathcal{IW}(Q_r^*, \alpha_r^*), \tag{23}$$

where

$$Q_r^* = Q_{0,r} + E_r' E_r, \tag{24}$$

$$E_r = Y_r - Z_r \Phi_r, \tag{25}$$

and

$$\alpha_r^* = \alpha_{0,r} + T_r. \tag{26}$$

These conditional posterior distributions are evaluated separately for the two regimes.

Gibbs Sampling Algorithm

The regime-specific OLS estimates are used as the initial values of Φ_r and Q_r. Because the threshold γ and delay lag d are fixed, observations remain assigned to the same regimes throughout the Gibbs sampler.

For iteration $m = 1, \dots, M$, the following steps are performed:

1. For each regime r, draw
$$\phi_r^{(m)} \sim p\left(\phi_r \mid Q_r^{(m-1)}, Y_r\right).$$
2. Reshape $\phi_r^{(m)}$ into the coefficient matrix $\Phi_r^{(m)}$.
3. Check the stability of the VAR polynomial in each regime. Draws that imply an unstable regime-specific VAR are discarded and redrawn.
4. Conditional on $\Phi_r^{(m)}$, draw
$$Q_r^{(m)} \sim p\left(Q_r \mid \Phi_r^{(m)}, Y_r\right).$$
5. After updating both regimes, use the retained parameter draws to calculate the impulse response functions.

The first `priors.burn` iterations are discarded as burn-in, and posterior inference is based on the subsequent `priors.keep` retained draws.

A Metropolis–Hastings step is not required because the threshold value is not estimated within the sampler. If the threshold were treated as an unknown parameter, an additional sampling step would be needed because its conditional posterior distribution is nonstandard.

Structural Shocks

For each regime, let P_r be an impact matrix satisfying

$$Q_r = P_r P_r'. \tag{27}$$

The reduced-form residuals can then be represented as

$$u_{r,t} = P_r \varepsilon_t, \qquad \varepsilon_t \sim N(0, I_N). \tag{28}$$

Under recursive identification, P_r is obtained from the lower-triangular Cholesky factorization of Q_r. With the ordering

$$\begin{bmatrix} g_t & \tau_t & y_t & \widetilde{CU}_t \end{bmatrix}', \tag{29}$$

government spending is ordered first, followed by net taxes, GDP, and capacity utilization.

Because the covariance matrices are allowed to differ,

$$Q_1 \neq Q_2, \tag{30}$$

the corresponding impact matrices may also differ:

$$P_1 \neq P_2. \tag{31}$$

The baseline analysis focuses on the responses to a positive government spending shock, although the same procedure can be applied to the other structural shocks.

Impulse Response Functions

Two types of impulse response functions are reported: fixed-state responses and evolving-state generalized impulse response functions (GIRFs). The GIRF measures the difference between a forecast path generated with a government spending shock and a baseline forecast path generated without the shock.

Let

$$\Theta^{(m)} = \left\{ \Phi_1^{(m)}, Q_1^{(m)}, \Phi_2^{(m)}, Q_2^{(m)} \right\}$$

denote posterior draw m of the regime-specific coefficients and covariance matrices, and let Ψ_{t-1} denote the history of the endogenous variables available before the shock. The GIRF at horizon h is defined as

$$GIRF(h, \Psi_{t-1}, \Theta^{(m)}) = E\left[Y_{t+h} \mid \varepsilon_t^g, \Psi_{t-1}, \Theta^{(m)} \right] - E\left[Y_{t+h} \mid \Psi_{t-1}, \Theta^{(m)} \right], \tag{32}$$

where ε_t^g denotes the structural government spending shock.

Simulation Procedure

For every initial history Ψ_{t-1} and posterior parameter draw $\Theta^{(m)}$, the GIRF is calculated using $B = 200$ simulations. The procedure is as follows:

1. Draw a sequence of future standardized disturbances for horizons $0, \ldots, H$.

2. Using the selected history and the simulated disturbances, generate a baseline forecast path without a government spending shock:

$$Y_{t+h}^{0,(m,b)}, \qquad h = 0, \ldots, H.$$

3. Using the same history and the same sequence of future standardized disturbances, generate a second forecast path after introducing a positive government spending shock in the initial period:

$$Y_{t+h}^{1,(m,b)}, \qquad h = 0, \ldots, H.$$

4. Calculate the difference between the shocked and baseline paths:

$$GIRF^{(m,b)}(h \mid \Psi_{t-1}) = Y_{t+h}^{1,(m,b)} - Y_{t+h}^{0,(m,b)}.$$

5. Repeat the simulation 200 times and average the responses:

$$\widehat{GIRF}^{(m)}(h \mid \Psi_{t-1}) = \frac{1}{200} \sum_{b=1}^{200} GIRF^{(m,b)}(h \mid \Psi_{t-1}).$$

Using the same future disturbances in the shocked and baseline simulations ensures that the difference between the two paths is driven by the government spending shock rather than by different realizations of future disturbances.

The procedure is repeated for each relevant history and each retained Gibbs draw. Responses for a given initial regime are obtained by averaging over all histories that begin in that regime:

$$\widehat{GIRF}_r^{(m)}(h) = \frac{1}{|\mathcal{H}_r|} \sum_{\Psi_{t-1} \in \mathcal{H}_r} \widehat{GIRF}^{(m)}(h \mid \Psi_{t-1}), \tag{33}$$

where $\mathcal{H}_r$ is the set of histories beginning in regime r. Posterior point estimates and credible intervals are then calculated from the distribution of the GIRFs across the retained Gibbs draws.

Fixed-State Responses

Fixed-state responses assume that the economy remains in its initial regime over the entire response horizon. A response beginning in the low-utilization state is generated using Φ_1 and Q_1 at every horizon, while a response beginning in the high-utilization state is generated using Φ_2 and Q_2 at every horizon.

Thus, future movements in capacity utilization do not cause a change in the regime used to generate the response. These responses isolate the propagation of the government spending shock within each regime.

Fixed-state responses provide a direct comparison of the dynamics within each regime.

Evolving-State Responses

Evolving-state responses allow the economy to move between the low- and high-utilization regimes after the shock. At each horizon, lagged mean-adjusted capacity utilization is compared with the threshold:

$$r_{t+h} = \begin{cases} 1, & \widetilde{CU}_{t+h-d} \leq \gamma, \\ 2, & \widetilde{CU}_{t+h-d} > \gamma. \end{cases} \tag{34}$$

The coefficients and covariance matrix used to generate the next observation are selected according to the resulting regime:

$$\left(\Phi_{r_{t+h}}, Q_{r_{t+h}}\right). \tag{35}$$

The regime sequence is determined separately for the shocked and baseline paths because the government spending shock may change the path of capacity utilization. Consequently, the shocked and baseline economies may switch regimes at different dates.

Because Q_1 and Q_2 may differ, the common standardized disturbances are transformed using the covariance matrix corresponding to the regime of each simulated path. Evolving-state responses therefore account for both regime-specific dynamics and endogenous transitions between regimes.

Interpreting the GIRFs

Real government spending, real net taxes, and real GDP enter the model as 100 times their first log differences. The raw GIRF for these variables is therefore measured in percentage points of quarterly growth. For example, a GDP response of 0.20 indicates that GDP growth is approximately 0.20 percentage points above its baseline value at that horizon.

The raw growth-rate responses are not yet dollar-for-dollar responses or fiscal multipliers. They must first be accumulated and converted into level responses.

Cumulative Level Response

For a variable X, the accumulated growth response through horizon h is

$$C_X(h) = \sum_{j=0}^{h} GIRF_X(j). \tag{36}$$

Because the model is expressed in log differences, $C_X(h)$ approximates the percentage deviation of the level of X from its baseline level at horizon h:

$$100\left[\log X_{t+h}^1 - \log X_{t+h}^0\right] \approx C_X(h). \tag{37}$$

Conversion into Dollar Responses

Let X_t^{level} denote the historical real-dollar level associated with the initial history. The approximate dollar response at horizon h is

$$\Delta X_{t+h}^{\$} \approx X_t^{\text{level}} \frac{C_X(h)}{100}. \tag{38}$$

In the implementation, the corresponding historical level observations are stored in the level-data matrix. This conversion is carried out separately for each initial history because the dollar levels of government spending and GDP differ across dates.

Shock Normalization

The initial government spending shock is normalized to equal 1% of GDP. Its dollar value for a particular history is therefore

$$Shock_t^{\$} = 0.01\, Y_t^{\text{level}}. \tag{39}$$

A dollar-for-dollar output response relative to the initial shock can be reported as

$$R_Y(h) = \frac{\Delta Y_{t+h}^{\$}}{Shock_t^{\$}}. \tag{40}$$

This statistic measures the dollar change in GDP at horizon h for each dollar of the initial government spending shock.

Cumulative Government Spending Multiplier

The cumulative fiscal multiplier through horizon H is calculated as the ratio of the cumulative dollar change in GDP to the cumulative dollar change in government spending:

$$M(H) = \frac{\displaystyle\sum_{h=0}^{H} \Delta Y_{t+h}^{\$}}{\displaystyle\sum_{h=0}^{H} \Delta G_{t+h}^{\$}}. \tag{41}$$

The denominator includes the full endogenous path of government spending following the initial shock, rather than only the dollar value of the impact-period shock. The multiplier therefore measures the cumulative increase in GDP associated with one cumulative dollar of additional government spending over the specified horizon.

References

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